Thomas Derrien

PhD Student in Computer Science, Université de Rennes, IRISA

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Research Interests

Multimodal entity linking; deep learning; NLP; computer vision; retrieval; dataset construction for evaluation.

Education

2025 – present **PhD in Computer Science**
CNRS, INSA Rennes, IRISA, Archimedia team, Rennes, France
- Thesis: *Multimodal Entity Linking*
- Advisors: Pascale Sébillot and Laurent Amsaleg
- Model training and evaluation
- Construction of evaluation datasets
- Presented work at SIGIR 2026 (Melbourne, Australia)

2022 – 2025 **Diplôme d'Ingénieur in Computer Science (MSc-level)**
INSA Rennes, Rennes, France
- Specialization in data science
- Mathematics, theoretical computer science, software engineering
- Exchange semester (Fall 2024): MSc Data Science, Berliner Hochschule für Technik, Berlin, Germany

2020 – 2022 **BSc in Mathematics and Applications**
Université de Rennes, Rennes, France
- Algebra, analysis, probability, applied mathematics
- Computer science minor (Python, Java, object-oriented programming)

2017 – 2020 **Baccalauréat Scientifique, with highest honours**
Lycée Tristan Corbière, Morlaix, France

Publications

[1] **T. Derrien**, L. Amsaleg, P. Sébillot. FAST-MEL: A Fast, Accurate, and Storage Efficient Solution for Multimodal Entity Linking. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26)*, Melbourne, Australia, 2026. doi:10.1145/3805712.3809860

Research Experience

Feb. – Aug. 2025 **Research Intern – Multimodal Entity Linking**
Inria, IRISA, Archimedia team, Rennes, France
- Literature review and study of existing approaches
- Critical analysis of methods and identification of promising research directions
- Study of a dense candidate-generation strategy
- Study of a multi-image approach to entity representation

Jun. – Jul. 2023 **Data Analysis Intern**
CNRS, Institut des Neurosciences Cellulaires et Intégratives
- Analyzed neuroscience data; built a Python tool to characterize responses to optogenetic stimulation
- Results and analyses used in a PhD thesis: https://theses.hal.science/tel-04892874

Other Professional Experience

Jun. – Sep. 2024 **Systems Engineering Intern**
French Ministry of the Interior, Digital Transformation Directorate
- Comparative study, deployment and improvement of an IT inventory tool for ∼1,000 servers running the ministry's messaging infrastructure
- DevOps automation pipelines and internal tooling (Python, Bash)
- Environment: GitLab CI/CD, Docker, Ansible

Technical Skills

Programming	Python, Java, Bash
ML & data	PyTorch, Hugging Face Transformers, Weights & Biases, scikit-learn, NumPy, pandas
Tools	Git, GitLab CI/CD, Docker, Ansible, LaTeX